Carlo M. Vilches

Research Economist — Energy Markets, Regulation & Quantitative Modeling

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Spanish nationality · mobile

PROFILE

Economist (10+ years) specialised in sectoral regulation, competition and tariff design, used to defending strategic interests before regulatory authorities. I build economic models for forecasting and impact assessment, cost the compensation of public service obligations and subsidy schemes, and communicate results clearly to senior management and institutional decision-makers. Proficient in statistical and econometric tools (Python, R) and in Excel. Fluent French (C1) and English (C1).

SKILLS & METHODS

Languages (prog.): Python · R · SQL · SAS · Stata

Quantitative: econometrics · modelling & forecasting · impact assessment · cost-benefit analysis

Languages: French C1 · Spanish (native) · English C1 · Catalan (basic)

Office / BI: Excel (advanced) · PowerPoint · Power BI

Specialisation: sectoral regulation · competition · public service obligations · tariff & cost allocation · compensation / subsidies

EXPERIENCE

Senior Analyst — Regulation & Competition · Osinergmin (energy regulator, Peru) 2019–2024

- Defended interests in 50+ competition disputes: quantified economic analysis, market power and valuation.
- Quantitative analysis and economic modelling for formal proceedings before regulatory bodies.
- Pedagogical communication of results to senior management and technical committees.

Analyst — Policy & Economic Analysis · Osinergmin · Directorate of Policy & Economic Analysis 2014–2019

- Economic costing and impact evaluation of public service obligations and their compensation (FOSE and MCTER subsidy schemes).
- Tariff design: rate structure, cost allocation and revenue requirements, linked to regulatory accounting.
- Economic models for forecasting and impact studies; international benchmarking.

Macroeconomic Policy Analyst · Ministry of Economy & Finance (MEF), Peru 2012–2013

- Macroeconomic modelling and forecasting (VAR models) in support of public decision-making.

EDUCATION

MSc Economics, Regulation & Competition — Universitat de Barcelona (2025–2026)
MSc Data Science — La Salle – URL, Barcelona (2025–2026)
MSc Behavioral Economics — Paris School of Economics (2018)
MSc Industrial Economics — Toulouse School of Economics (2005–2007)
BSc Economics — PUCP, Peru (2000–2004)

WORKING PAPERS

ELECTRICITY & ENERGY MARKETS

1. The Price of Hedging: Forward Risk Premia in Iberian Power through the Energy Crisis, and What They Cost the Non-Integrated Retailer (2019–2024).
   github.com/cmvdata/mibel-derivatives/blob/main/paper_derivatives/derivatives_paper.pdf
2. Capacity Market Design and Renewable Investment in Spain: A Forensic Audit of the Derating Factors of the 2026 Capacity Mechanism (SA.112483 / RD 932/2024) Using Hourly ELCC Data (2019–2024).
   github.com/cmvdata/capacity/blob/main/paper/cm_spain_paper.pdf

3. Exit and Absorption in Spanish Electricity Retail through the Energy Crisis (2019–2024).
   github.com/cmvdata/model_electricity/blob/main/retail_paper.pdf
4. No Firm Capacity, Three Markets: Solar and the Capacity Mechanism in Southern Europe.
   github.com/cmvdata/southern-europe-renewables/blob/main/paper/three_markets.pdf

TRANSPORT

1. The Premium-Calibration Sensitivity of an Intermodal Pricing Ceiling: A Theory-Grounded Yield-Management Engine for the BCN–MAD Corridor.
   github.com/cmvdata/yield-management-bcn-mad/blob/main/paper/yield_bcn_mad_paper.pdf
2. Where Did the Booking Fences Go? Advance-Purchase Pricing in a Post-Entry Air Duopoly on the Madrid–Barcelona Corridor.
   github.com/cmvdata/yield-management-bcn-mad/blob/main/paper_pricing/pricing_paper.pdf
3. A Corridor Loses Both of Its Low-Cost Operators: Prices, Passengers, and a Registered Prediction.
   github.com/cmvdata/yield-management-bcn-mad/blob/main/paper_avlo/avlo_preres.pdf
4. Pricing Response After the Low-Cost Brand Leaves: Independent High-Speed Fares on Madrid–Barcelona When AVLO Withdrew.
   github.com/cmvdata/yield-management-bcn-mad/blob/main/paper_railprice/railprice_paper.pdf
5. Subsidising the Metro per Passenger or per Kilometre? A Calibrated Translog Cost Function for the Barcelona Metro (FMB-TMB), 2023–2025.
   github.com/cmvdata/TMB-costos/blob/main/paper/paper_translog_fmb.pdf
6. The Generalised Price of the Madrid–Barcelona Corridor: What Realised Market Data Reveal About the Air–Rail Trade-off.
   github.com/cmvdata/yield-management-bcn-mad/blob/main/paper_genprice/genprice_paper.pdf

DATA SCIENCE / MACHINE LEARNING

1. Detecting Structural Manipulation in Electricity Markets: A Supervised Approach Using Partial Regulatory Ground Truth.
   github.com/cmvdata/neuro/blob/main/paper/main.pdf
2. Replicating LEAR on MIBEL through the Solar Transition: Why Technical Indicators Do Not Transfer, and Why Percentage Errors Mislead (2022–2024).
   github.com/cmvdata/mibel-forecasting/blob/main/paper_forecasting/forecasting_paper.pdf
3. When Psychopathic Traits Are Not Traits: Aggregation Bias and Big-Five Redundancy in the Prediction of Professional Success.
   github.com/cmvdata/psychopathy-success/blob/main/docs/paper.pdf
4. Cross-Sectional Characterization of High-Cost Healthcare Users in the US: How the Morbidity-Cost Gradient Varies by Insurance Type in MEPS 2022.
   github.com/cmvdata/MEPS_analysis/blob/main/paper/meps_paper.pdf

PUBLIC ECONOMICS · HEALTH · PSYCHOMETRICS

1. Energy Poverty and the Targeting of Electricity Subsidies: Evidence from the FOSE in Peru. *(Master's thesis)*
   github.com/cmvdata/fose_analysis/blob/main/docs/paper_fose.pdf
2. Targeting Energy Poverty: Structural Predictors versus Income Thresholds in Spain's Social Electricity Bonus.
   github.com/cmvdata/bono-social-microsim/blob/main/paper/paper.pdf
3. Ordinal Distortion of the Spanish Regional Financing System: Quantitative Evidence for the Common-Regime Autonomous Communities, 2009–2022.
   github.com/cmvdata/sfa-distorsion-ordinal/blob/main/paper/paper.pdf

PUBLISHED & INSTITUTIONAL

Co-editor of an Osinergmin volume on the Peruvian electricity industry (2016) and co-author of two working papers on energy-sector regulation (Working Papers 32 and 40, 2014–2016). References available on request.